

A neuron is a weighted vote

ONE NEURON DECIDES: LEND OR NOT?

a small loan application



multiply each input by its weight $\rightarrow$ add them up $\rightarrow$ cross the line or not

Learning is only this: nudging those weights until the answers come out right

Multiply, add, decide

One neuron does three things: **multiply, add, decide**. That is it.

- 1 Each input is multiplied by a **weight** — how much that input matters.
- 2 The results are added together into a single number.
- 3 If that number crosses a threshold, the neuron **fires**.

REMEMBER

Learning is nothing more than adjusting those weights until the votes come out right.

